RESUMIND

AI Resume Analyzer

Smart, honest feedback for your dream job — ATS score, Europass CV and cover letter

Scan to try it live

ai-resume-analyzer-psi-lime.vercel.app

CSE4204 Mobile Computing Lab · Team CSE4204-8A-T05 · Section 8A

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GitHub: github.com/mahamud894/AI_Resume_Analyzer

Supervisor: Md. Riaz Mahmud, Assistant Professor, Dept. of CSE, NUBTK

Good CVs are rejected before a human reads them

Filtered by ATS

Applicant Tracking Systems reject resumes with missing keywords, weak verbs or unclear structure — automatically.

No feedback

Applicants get repeated rejections without knowing what to fix.

Unknown standard

International employers expect a Europass CV (MM/YYYY dates, CEFR, EQF) that most graduates were never taught.

Time-consuming

A personalised cover letter for every application takes hours, so most people send generic ones.

Existing tools fall short: Jobscan and Resume Worded are paid after a few scans and match keywords without context; Reactive Resume is only a builder with no AI analysis. None combine analysis, a Europass rewrite and a cover letter for the same job.

Objectives

O1

Secure accounts

bcrypt + JWT, Google Sign-In

O2

Document vault

PDFs and history in MongoDB

O3

AI ATS analysis

Score + 5 categories with tips

O4

Europass rewrite

Fix every tip, never invent facts

O5

Before / after

Measure the gain with a new score

O6

Cover letter

One page, tied to the same job

O7

Any device

Responsive, mobile-first UI

O8

Quality & deployment

48 tests, CI, live on the web

One flow from upload to cover letter

1. Upload

PDF + job title and description

2. Analyse

Overall score, ATS, tone, content, structure, skills

3. Europass CV

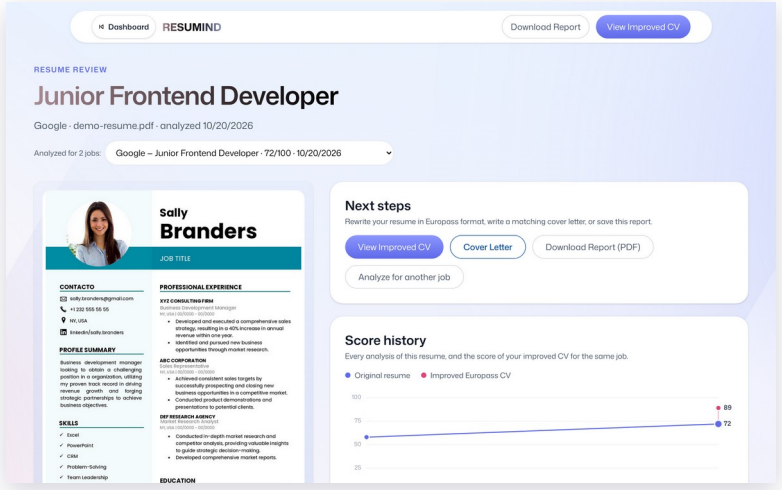
Rewrite, inline edit, change list, PDF

4. Before / after

Same job, new score, history chart

5. Cover letter

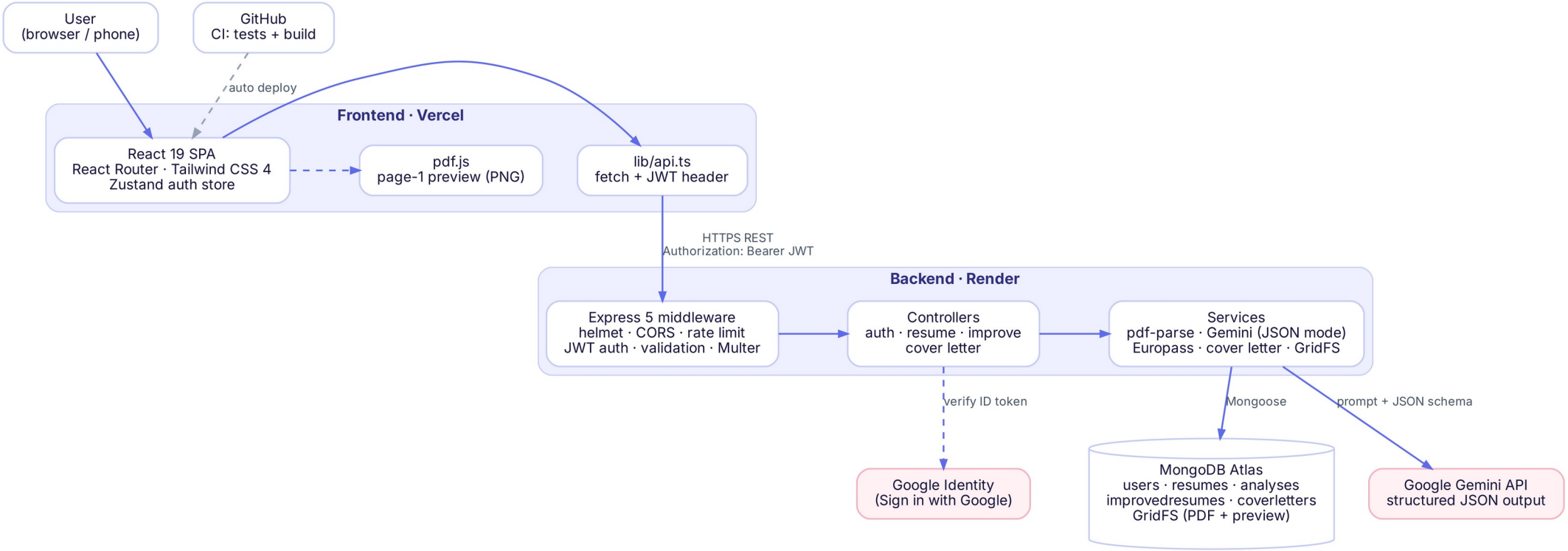
Formal or modern, editable, PDF



Also included

- History dashboard and printable report
- Email/password and Google Sign-In
- Change password, delete account
- Phone, tablet and desktop layouts

Architecture: three tiers, secrets only on the server

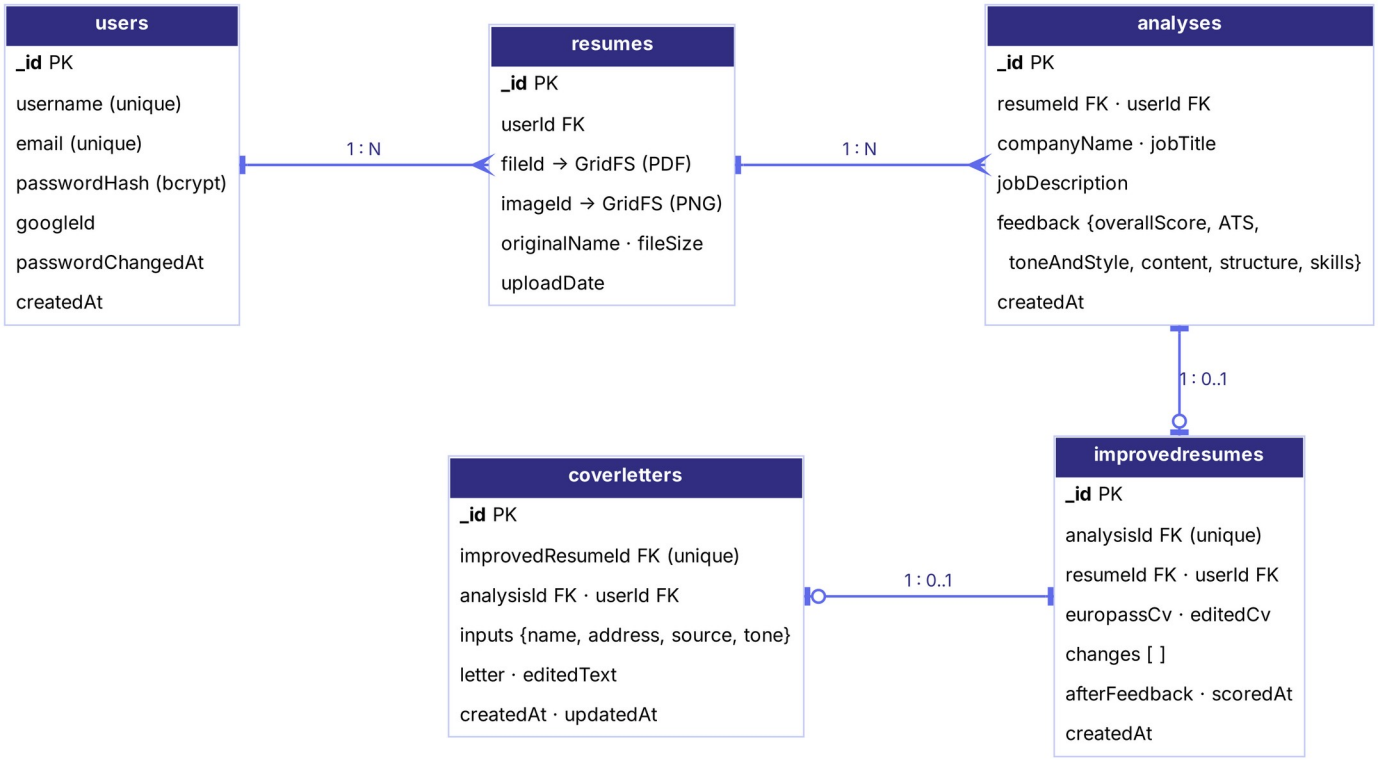


Frontend · Vercel
React 19, React Router 8, Tailwind CSS 4, Zustand, pdf.js

Backend · Render
Node.js, Express 5, Multer, pdf-parse, helmet, express-validator

Data & AI
MongoDB Atlas + GridFS, Google Gemini, Google Identity

Database design



users 1–N resumes 1–N analyses 1–0..1 improvedresumes 1–0..1 coverletters

- userId on every document — queries are always filtered by owner
- Unique indexes: one CV per analysis, one letter per CV
- Cascade delete for resume and account
- AI version and user edits stored side by side
- PDFs and previews in GridFS — safe across redeploys

AI workflow

pdf-parse

Extract text (422 if scanned)

Prompt

Resume + job + rules

Gemini

JSON mode + schema

Retry

Busy → wait 2 s, then 4 s

Normalise

Clamp scores, drop bad tips

Save

Analysis → result page

Analysis
Overall score + ATS, tone & style, content, structure, skills — each with good / improve tips

Europass rewrite
Every "improve" tip + job → Europass CV and a list of changes

Before / after
Improved CV → text, placeholders removed → same prompt, same job

Cover letter
AI writes opening, body, closing; code adds header, subject, salutation, sign-off

Keeping the AI honest

3×

automatic retries when Gemini is busy

15/h

AI requests per user

0

invented numbers allowed in the CV

postCheck() in code fixes what the model gets wrong

AI wrote	After post-check
improved load time by 40%	by [Add a measurable result, e.g. X%]
November 30, 2024	11/2024
Current (ongoing degree)	Expected 08/2026
H.S.C, 4.67	Higher Secondary Certificate (HSC), GPA 4.67 / 5.00, EQF 4
Religion, date of birth	removed (not part of Europass)

Placeholders show the user exactly what to fill in — the CV stays truthful.

Authentication, security and API

bcrypt + JWT
Passwords 8–72 chars, hashed. Token valid 7 days; password change logs out other devices.

Google Sign-In
ID token verified on the server; only verified e-mails.

Hardening
helmet, CORS allow-list, validation on every input, 404 for other users' data.

Rate limits
20 auth requests / 15 min per IP; 15 AI requests / hour per user.

21 REST endpoints

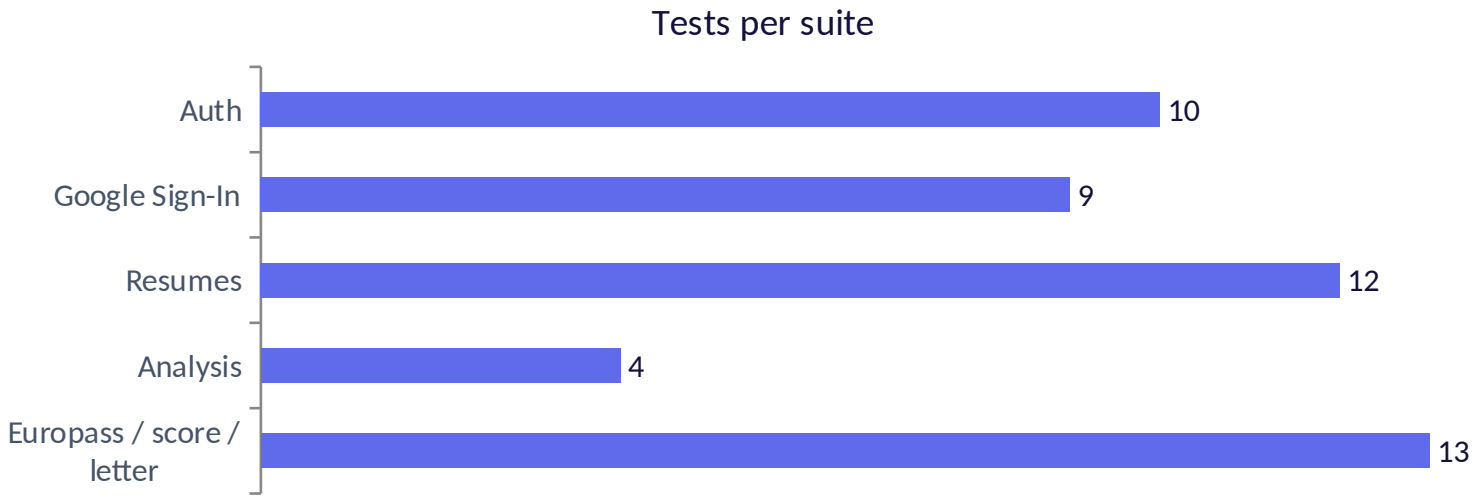
Group	Endpoints
Auth (6)	register, login, google, me, password, delete
Resume (6)	upload, list, get, file, image, delete
AI (8)	analyze, improve, get/edit CV, score, cover letter get/create/edit
Health (1)	/api/health for Render

```
Authorization: Bearer <JWT>  ·  JSON errors
{ message, details }
```

Testing and continuous integration

48

automated API tests, all passing in GitHub Actions



Vitest + Supertest

Real HTTP requests against the Express app

In-memory MongoDB

Fresh database every run; no real data touched

Gemini mocked

Fast, repeatable, no API key needed

CI on every push

Backend tests + frontend type check and build

Deployment: live on the web

Vercel Frontend · static SPA

https://ai-resume-analyzer-psi-lime.vercel.app

Render Backend · render.yaml Blueprint

resumind-api-3kpd.onrender.com

MongoDB Atlas Database + GridFS files

M0 cluster

GitHub Actions Tests before every deploy

github.com/mahamud894/AI_Resume_Analyzer

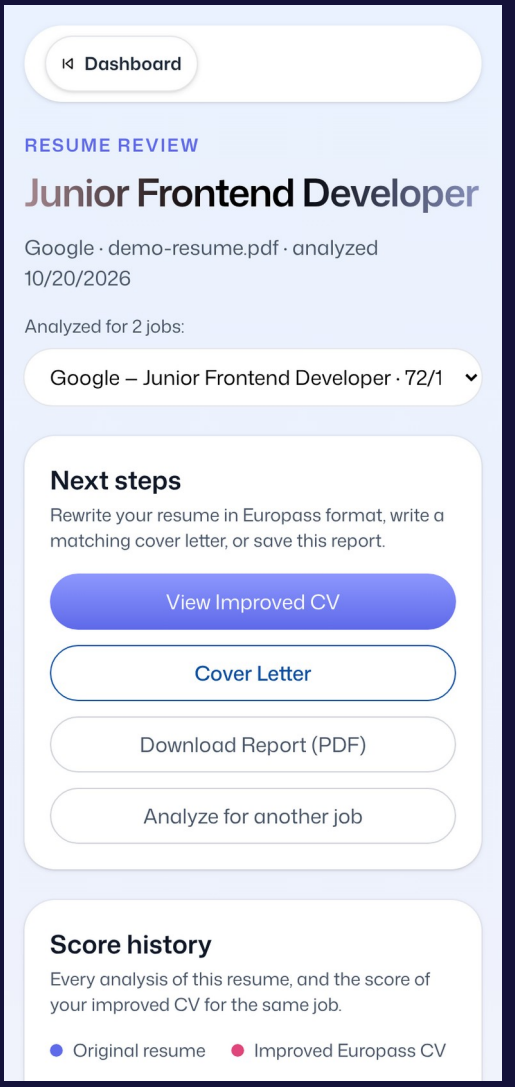


Try it now

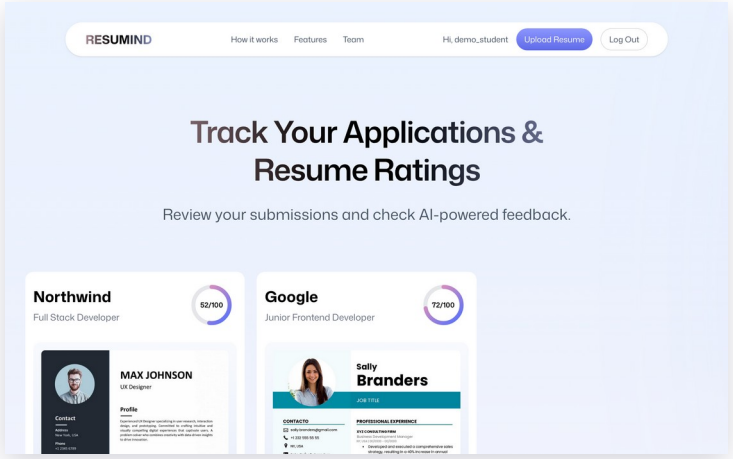
Secrets live only in .env and the Render / Vercel dashboards — never in Git.

Live demonstration

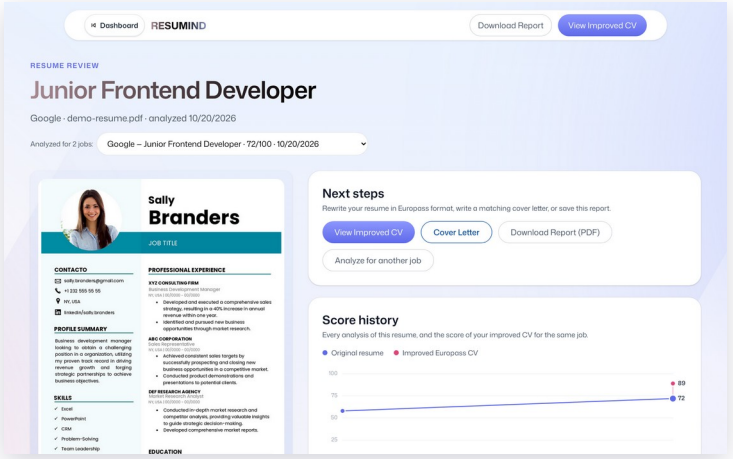
1. Open the live site on laptop and phone
2. Sign up / Sign in with Google
3. Upload a resume for a real job
4. Read the scores and tips
5. Improve to Europass, edit a line
6. Score the improved CV
7. Generate and download the cover letter
8. Dashboard history + green CI



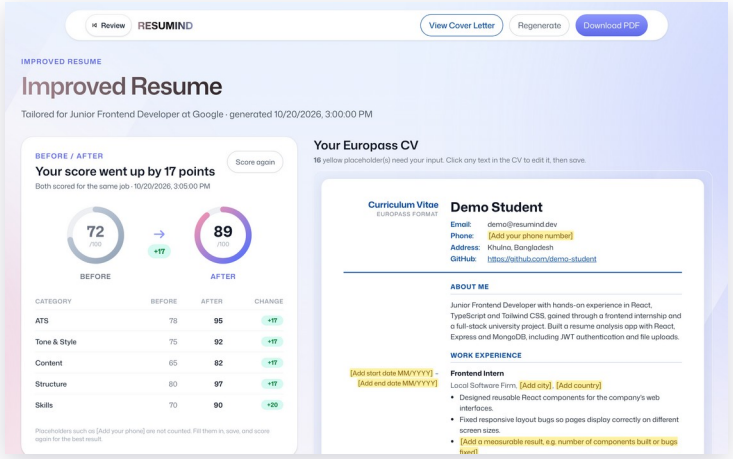
Screens (backup for the demo)



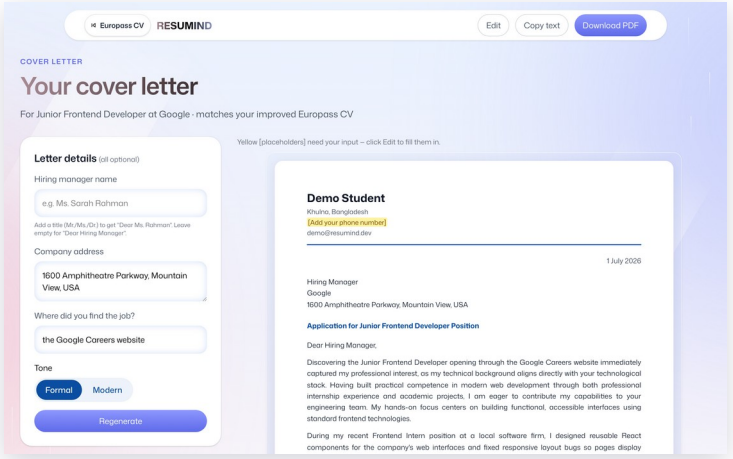
Dashboard



Analysis result



Europass CV



Cover letter

Challenges and how we solved them

Inconsistent AI output

→ JSON mode + strict schema + validation in code

AI invented numbers and dates

→ Deterministic post-check with placeholders

Gemini "high demand" errors

→ Retries with back-off, clear messages

Files lost on redeploy

→ Moved storage to MongoDB GridFS

One rate limit for all users behind a proxy

→ TRUST_PROXY so each real IP counts

Document-heavy pages on phones

→ Mobile-first layout, A4 sheets that scale

Future improvements

OCR for scanned PDFs

Read image-only resumes and DOCX files

Job link import

Paste a LinkedIn or BDJobs link instead of the description

Interview practice

AI questions based on the CV and the job

Bangla support

Interface and feedback in Bangla

Native mobile app

React Native on the same REST API, push notifications

Recruiter mode

Rank many resumes against one job

Thank you — questions?



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Team Leader · Frontend + UI/UX



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Core System Integration



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Database & Quality



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